CS353 Computer Networks Lab — Assignment 10

Network Layer: IP Routing Table & Longest Prefix Match (LPM) Manual Solution

1. Fundamental Concepts & Lookup Rules

• **CIDR Prefix & Subnet Mask:** An IPv4 address is 32 bits. A prefix /n indicates that the leading **n bits** represent the Network ID (fixed), while the remaining **(32 - n) bits** represent Host IDs.
• **Bitwise Matching Condition:** A destination IP address D matches a routing table entry with network N and mask M if and only if: **(D & M) == N** [where **&** denotes bitwise AND].
• **Longest Prefix Match (LPM) Principle:** When an IP matches multiple overlapping routes (e.g. /16, /22, /25), the router selects the entry with the **highest prefix length (largest /n)** — representing the most specific route.
• **Default Route (0.0.0.0/0):** Prefix length /0 (mask 0.0.0.0). Matches every IPv4 address as a universal fallback.

2. Analysis of the Given Routing Table (Table 1)

Entry	Destination Network	Prefix	Subnet Mask	Subnet Address Range	Next Hop
E1	`0.0.0.0`	/0	`0.0.0.0`	`0.0.0.0 – 255.255.255.255`	198.51.100.1 (default route)
E2	`172.20.0.0`	/16	`255.255.0.0`	`172.20.0.0 – 172.20.255.255`	10.10.10.2
E3	`172.20.4.0`	/22	`255.255.252.0`	`172.20.4.0 – 172.20.7.255`	10.10.10.3
E4	`172.20.4.128`	/25	`255.255.255.128`	`172.20.4.128 – 172.20.4.255`	10.10.10.4
E5	`192.0.2.0`	/24	`255.255.255.0`	`192.0.2.0 – 192.0.2.255`	directly connected (eth1)

Mathematical Range Derivation:
• **Entry E3 (172.20.4.0/22):** Mask /22 has 22 network bits (8+8+6). 3rd octet mask is 11111100_2 (252). In 3rd octet, network base 4 is 00000100_2. The 2 host bits range from 00_2 to 11_2 (values 4 to 7). Thus the 3rd octet ranges from **4 to 7**, giving 172.20.4.0 to 172.20.7.255.
• **Entry E4 (172.20.4.128/25):** Mask /25 has 25 network bits (8+8+8+1). 4th octet mask is 10000000_2 (128). The top bit is fixed to 1 (128). The remaining 7 host bits range from 0 to 127. Thus the 4th octet spans **128 to 255**, giving 172.20.4.128 to 172.20.4.255.

3. Step-by-Step Manual Lookup for Each Destination IP

a) Destination IP: 172.20.4.200

• 0.0.0.0/0: **MATCH** (Default route matches all IPs)
• 172.20.0.0/16: **MATCH** (Prefix 172.20 matches)
• 172.20.4.0/22: **MATCH** (3rd octet is 4, in range [4, 7])
• 172.20.4.128/25: **MATCH** (4th octet is 200 ≥ 128; bitwise 200 & 128 = 128)
• 192.0.2.0/24: **NO MATCH** (First octet 172 ≠ 192)
→ **All matching entries:** 0.0.0.0/0, 172.20.0.0/16, 172.20.4.0/22, 172.20.4.128/25
→ **Longest Prefix Match (LPM): 172.20.4.128/25** (Longest prefix: **/25**) • **Next Hop: 10.10.10.4**

b) Destination IP: 172.20.4.5

• 0.0.0.0/0: **MATCH** (Default route matches all IPs)
• 172.20.0.0/16: **MATCH** (Prefix 172.20 matches)
• 172.20.4.0/22: **MATCH** (3rd octet is 4, in range [4, 7])
• 172.20.4.128/25: **NO MATCH** (4th octet is 5 < 128; top bit is 0, required 1)
• 192.0.2.0/24: **NO MATCH** (First octet 172 ≠ 192)
→ **All matching entries:** 0.0.0.0/0, 172.20.0.0/16, 172.20.4.0/22
→ **Longest Prefix Match (LPM): 172.20.4.0/22** (Longest prefix: **/22**) • **Next Hop: 10.10.10.3**

c) Destination IP: 172.20.9.9

• 0.0.0.0/0: **MATCH** (Default route matches all IPs)
• 172.20.0.0/16: **MATCH** (Prefix 172.20 matches)
• 172.20.4.0/22: **NO MATCH** (3rd octet is 9 ∉ [4, 7]; 9 & 252 = 8 ≠ 4)
• 172.20.4.128/25: **NO MATCH** (3rd octet is 9 ≠ 4)
• 192.0.2.0/24: **NO MATCH** (First octet 172 ≠ 192)
→ **All matching entries:** 0.0.0.0/0, 172.20.0.0/16
→ **Longest Prefix Match (LPM): 172.20.0.0/16** (Longest prefix: **/16**) • **Next Hop: 10.10.10.2**

d) Destination IP: 192.0.2.77

• 0.0.0.0/0: **MATCH** (Default route matches all IPs)
• 172.20.0.0/16: **NO MATCH** | 172.20.4.0/22: **NO MATCH** | 172.20.4.128/25: **NO MATCH**
• 192.0.2.0/24: **MATCH** (First three octets 192.0.2 match exactly)
→ **All matching entries:** 0.0.0.0/0, 192.0.2.0/24
→ **Longest Prefix Match (LPM): 192.0.2.0/24** (Longest prefix: **/24**) • **Next Hop: directly connected (eth1)**

e) Destination IP: 203.0.113.50

• 0.0.0.0/0: **MATCH** (Default route matches all IPs)
• All other entries (172.20.0.0/16, 172.20.4.0/22, 172.20.4.128/25, 192.0.2.0/24): **NO MATCH**
→ **All matching entries:** 0.0.0.0/0
→ **Longest Prefix Match (LPM): 0.0.0.0/0** (Default route /0) • **Next Hop: 198.51.100.1 (default route)**

4. Consolidated Results Summary Table

Destination IP	Every Matching Entry	Selected Entry (LPM)	Resulting Next Hop
a) 172.20.4.200	`0.0.0.0/0, 172.20.0.0/16, 172.20.4.0/22, 172.20.4.128/25`	**172.20.4.128/25** (/25)	**10.10.10.4**
b) 172.20.4.5	`0.0.0.0/0, 172.20.0.0/16, 172.20.4.0/22`	**172.20.4.0/22** (/22)	**10.10.10.3**
c) 172.20.9.9	`0.0.0.0/0, 172.20.0.0/16`	**172.20.0.0/16** (/16)	**10.10.10.2**
d) 192.0.2.77	`0.0.0.0/0, 192.0.2.0/24`	**192.0.2.0/24** (/24)	**directly connected (eth1)**
e) 203.0.113.50	`0.0.0.0/0`	**0.0.0.0/0** (/0)	**198.51.100.1 (default route)**